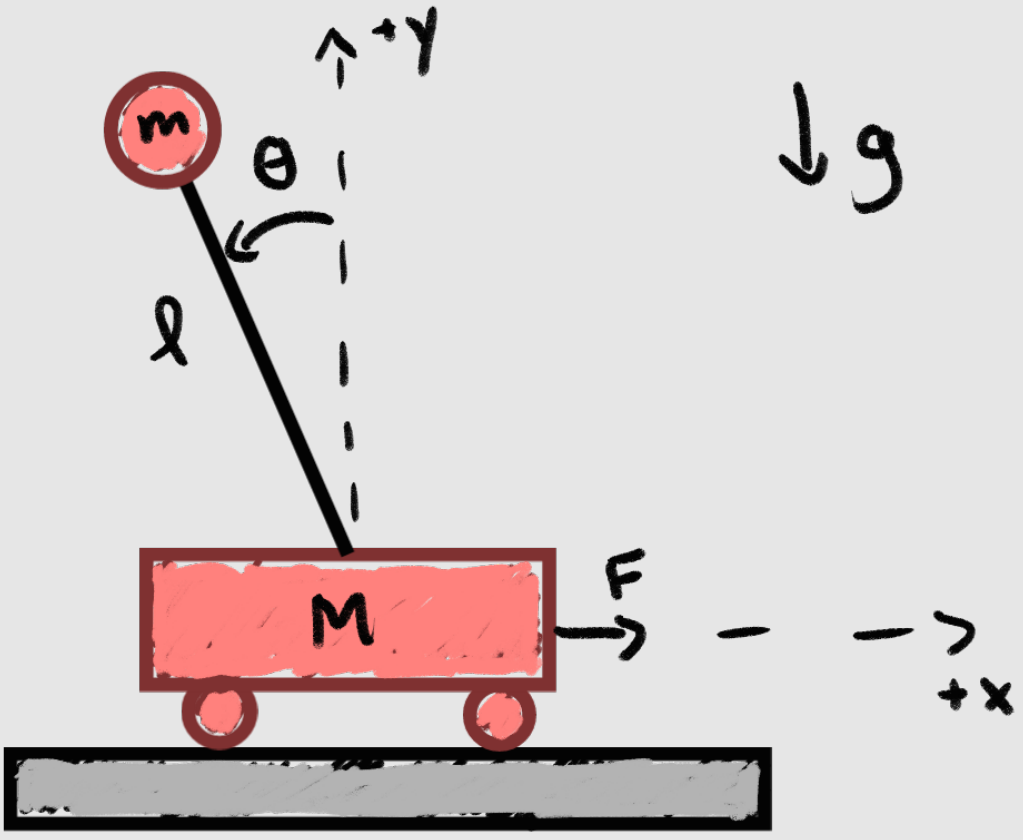
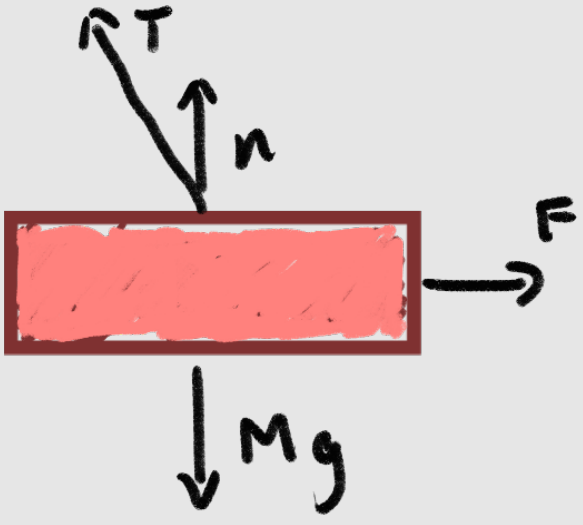

+y
m
θ
l
g
M
F
+x

NEWTONIAN APPROACH

KINEMATICS

FBD (cart)



$$\Sigma F_x : \quad F - T\sin\theta = M\ddot{x}$$

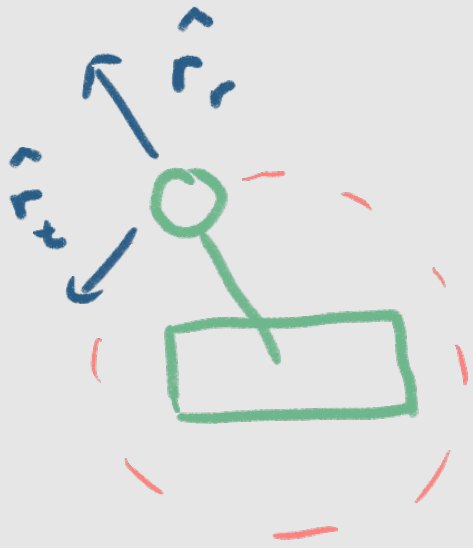
FBD (pendulum)



a_p = acceleration of pendulum

$$\Sigma F_x = T\sin(\theta) = m a_{px}$$

$$\Sigma F_y = -T\cos(\theta) - mg = m a_{py}$$



$$a_p = a_c + a_{p/c}$$

$$= \ddot{x}\hat{i} + L\ddot{\theta}\,\hat{r}_t - L\dot{\theta}^2\,\hat{r}_r$$

$$\hat{r}_t = -\cos(\theta)\hat{i} - \sin(\theta)\hat{j}$$

$$\hat{r}_r = -\sin(\theta)\hat{i} + \cos(\theta)\hat{j}$$

$$a_p = \ddot{x}\hat{i} + L\ddot{\theta}\left[-\cos(\theta)\hat{i} - \sin(\theta)\hat{j}\right]$$

$$- L\dot{\theta}^2\left[-\sin(\theta)\hat{i} + \cos(\theta)\hat{j}\right]$$

Recall:

$$T\sin(\theta) = m a_{px}$$

$$= m\ddot{x} - mL\ddot{\theta}\cos(\theta) + mL\dot{\theta}^2\sin(\theta)$$

and...

$$-T\cos(\theta) - mg = m a_{py}$$

$$= -mL\ddot{\theta}\sin(\theta) - mL\dot{\theta}^2\cos(\theta)$$

Also,

$$F - T\sin\theta = M\ddot{x}$$

$$-mg\sin\theta = m\ddot{x}\cos(\theta) - mL\ddot{\theta}$$

$$F + mL\ddot{\theta}\cos(\theta) - mL\dot{\theta}^2\sin\theta = (m+M)\ddot{x}$$

Simplifying:

$$L\ddot{\theta} - \ddot{x}\cos\theta - g\sin\theta = 0$$

$$(M+m)\ddot{x} - mL\cos(\theta)\ddot{\theta} + mL\sin(\theta)\dot{\theta}^2 = F$$

$$\begin{bmatrix} M+m & -mL\cos(\theta) \\ -\cos(\theta) & L \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F - mL\sin(\theta)\dot{\theta}^2 \\ g\sin(\theta) \end{bmatrix}$$

$$x = \frac{D_x}{D} \qquad y = \frac{D_y}{D}$$

$$D_x = c_1 b_2 - b_1 c_2$$

$$D_y = a_1 c_2 - c_1 a_2$$

$$D = a_1 b_2 - b_1 a_2$$

$$\ddot{x} = \frac{F - mL\sin\theta\,\dot{\theta}^2 + mg\sin\theta\cos\theta}{M + m\sin^2\theta}$$

$$\ddot{\theta} = \frac{(M+m)g\sin\theta + F\cos\theta - mL\sin\theta\cos\theta\,\dot{\theta}^2}{(M+m\sin^2\theta)L}$$

$$\dot{x} = Ax + Bu$$

Linearize: $(\theta = 0, \dot{\theta} = 0)$

$\sin(\theta) \approx \theta$ $\qquad$ $\sin^2(\theta) \approx 0$

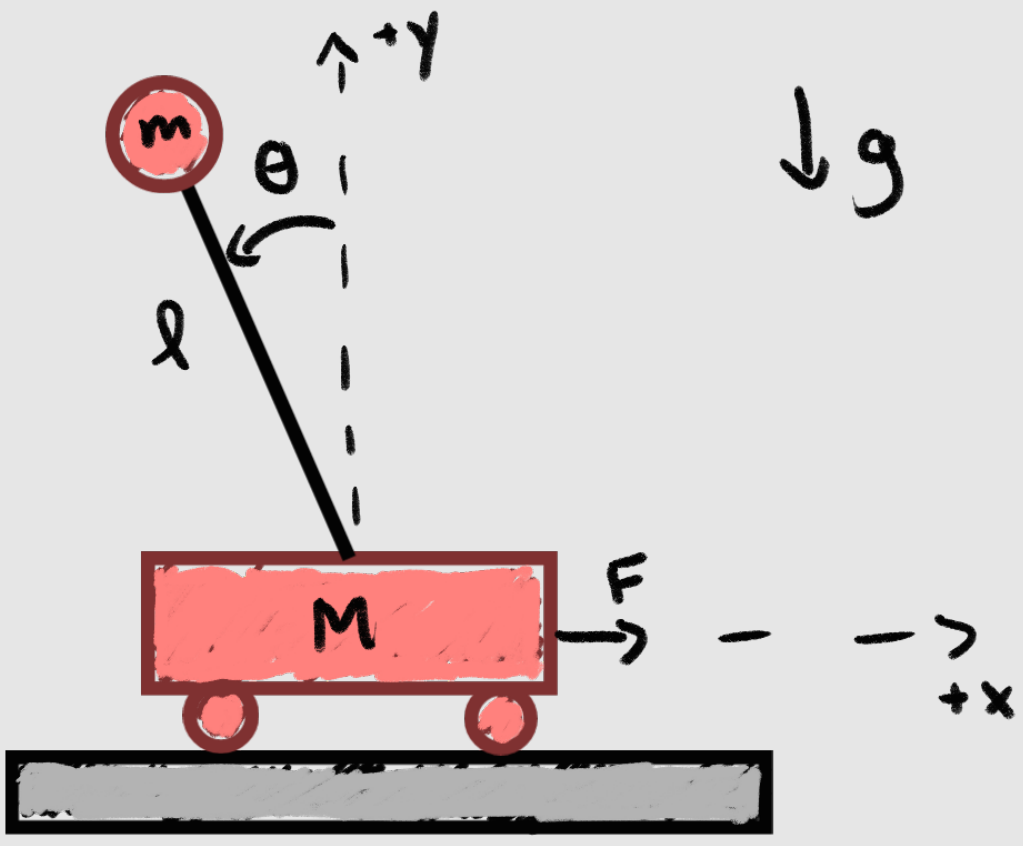
$\cos(\theta) \approx 1$ $\qquad$ $\dot{\theta}^2 \to 0$

For $\theta = \pi$ as equilibrium, switch signs

$$\ddot{x} \approx \frac{1}{M} F + \frac{mg}{M} \theta$$

$$\ddot{\theta} \approx \frac{1}{ML} F - \frac{(M+m)g}{ML} \theta$$

LAGRANGIAN APPROACH



$$x_m = x - L\sin\theta$$

$$y_m = L\cos\theta$$

$$\dot{x}_m = \dot{x} - L\dot{\theta}\cos\theta$$

$$\dot{y}_m = -L\dot{\theta}\sin\theta$$

Potential Energy

$$V = mgy_m = mgL\cos(\theta)$$

Kinetic Energy

$$T = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x}_m^2 + \dot{y}_m^2)$$

$$= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x}^2 - 2L\dot{\theta}\dot{x}\cos\theta - L^2\dot{\theta}^2)$$

$$T = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}mL^2\dot{\theta}^2 - mL\dot{\theta}\dot{x}\cos\theta$$

LAGRANGE'S EQUATIONS

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i$$

$$\mathcal{L} = T - V$$

$$\mathcal{L} = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 - mL\dot{\theta}\dot{x}\cos\theta - mgl\cos\theta$$

For x

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = Q_x$$

$$(M+m)\ddot{x} - (mL\ddot{\theta}\cos\theta - mL\dot{\theta}^2\sin\theta) = F$$

For θ

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = Q_\theta$$

$$mL^2\ddot{\theta} - mL\ddot{x}\cos\theta + mL\dot{x}\dot{\theta}\sin\theta - mL\dot{\theta}\dot{x}\sin\theta - mgL\sin\theta = 0$$

Simplify:

$$L\ddot{\theta} - \ddot{x}\cos\theta - g\sin\theta = 0$$

Writing w/ J and Δ

J: Moment of Inertia about the pivot

I_{cm} for uniform rod: $\frac{1}{12} m l_{rod}^2$

$$L = \frac{l_{rod}}{2}$$

I_{cm}: $\frac{1}{3} mL^2$

$$J = \frac{1}{3} mL^2 + mL^2 = \frac{4}{3} mL^2$$

Sub into where mL^2 was before

Δ: Determinant of new matrix w/ J evaluated at $\theta = 0$

For a sim w/ friction

Cart:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = F + Q_x$$

$$Q_x = -d\dot{x} - c_{d,cart}\dot{x}|\dot{x}|$$

(Linear viscous, Quadratic aero drag)

Pendulum:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial x} = Q_\theta$$

$$Q_\theta = -b\dot{\theta} - c_{d,pend}\dot{\theta}|\theta|$$